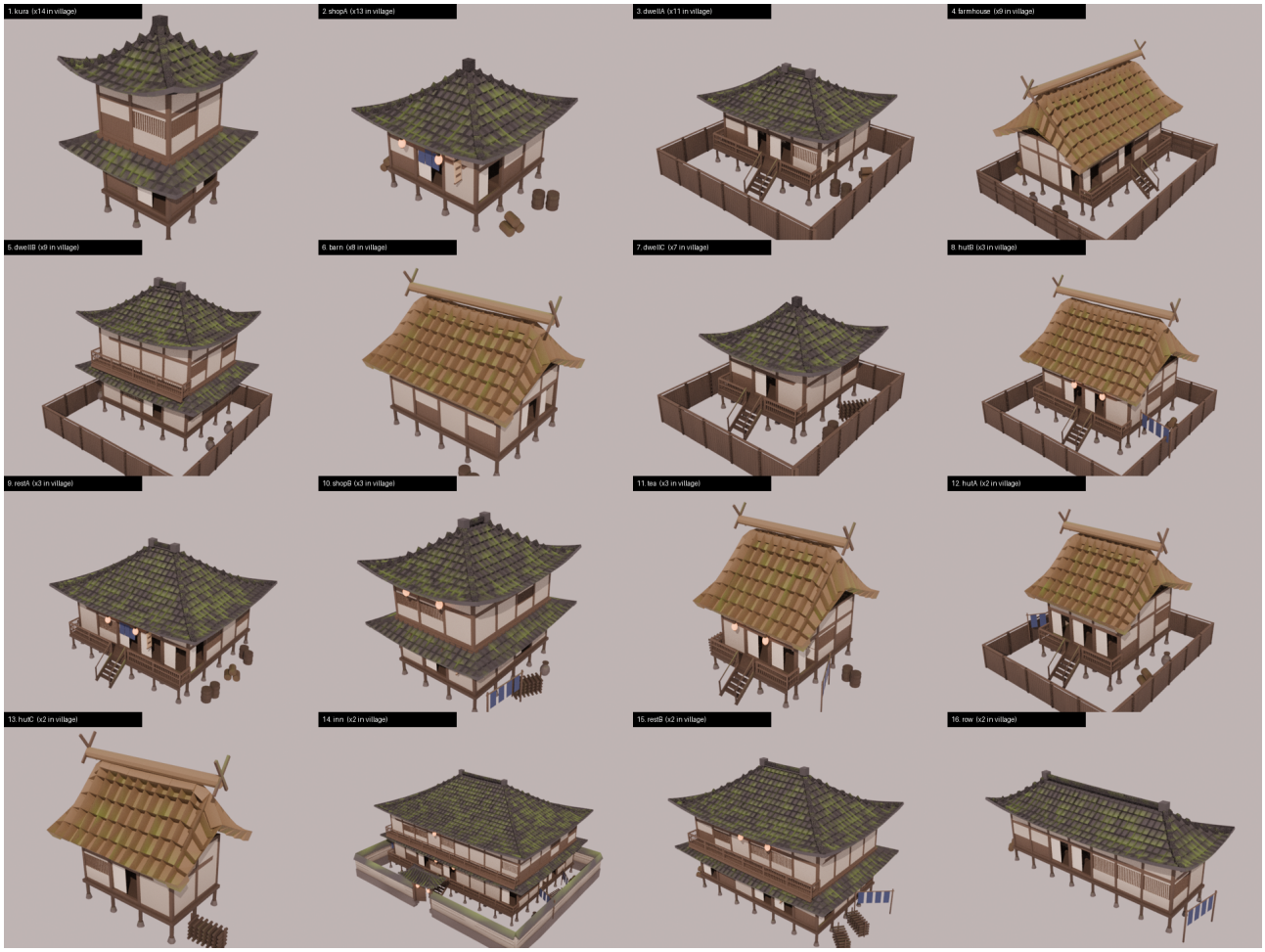
Edo Village House Kit

Deliverables specification: everything the finished result will contain

Source scene	Feudal_Japan4 - Copy.blend (Blender 5.2.2), 94 procedural houses, 17 designs
Target engine	Unreal Engine 5.8, PC, Nanite meshes, no LODs
Style	Edo-period Japanese village: tile hip roofs, thatched gable roofs, timber frame, plaster walls, stilted foundations
Quality goal	Game-ready, heavily optimized, as realistic as procedural + CC0 photo textures allow
Status	Analysis done. Building has not started. Nothing in your .blend has been changed.

How to read this document. Items marked **Agreed** were confirmed by you. Items marked **Proposed** are my defaults (you asked me to use my recommendations) and can still change. Section 9 lists what I can and cannot verify from this cloud container, and section 10 lists open risks. Nothing here has been built yet, so no figure for the finished kit is a measurement; figures about the existing scene are measured and come from the analysis report.



One house of each type, rendered from the source scene (numbers are how many of each exist in the village).

1. What you will receive

The result is a modular kit that rebuilds your village houses from a small set of reusable, grid-snapped pieces, with real textures and clean UVs, exported for Unreal Engine 5.8. It replaces the current box-based procedural geometry (unoptimized, no UVs, no textures) with properly modelled, texture-based pieces.

#	Deliverable	Format / location	Notes
1	Modular kit scene	.blend (Blender 5.2), repo folder *kit/*	All modules in one organized scene: collections per module family, named, origin-set, grid-snapped.
2	Per-module meshes	.fbx, one file per module, *export/fbx/*	Unreal-ready scale and axes, Nanite-friendly. No LODs. Collision added later.
3	Kit geometry nodes (new)	Node group *GN_EdoKitHouse* + helpers, inside the kit .blend	New geometry nodes built from the kit meshes from the start. Same style of inputs as your generator (Bays X/Y, Floors, Roof style, Shopfront, Engawa, Balcony, Lanterns, Gate, Fence, Props) plus a variant seed. Instances the kit modules. Your existing *GN_EdoHouse* is not modified.
4	Textures	PNG, *kit/textures/* (and *export/textures/*)	Shared 4K sets for the main materials, 2K for minor ones. BaseColor, Normal (DirectX), packed ORM, plus emissive where needed.
5	Assembly manifest	.json, *export/manifest.json*	For each of the 94 houses: which modules, where, rotation. Rebuilds the village from the kit.
6	Unreal import helper	.py script for the UE Python console	Imports FBX/textures and (optionally) places houses from the manifest. **Untested**: I have no Unreal here.
7	Preview renders	.png, *previews/*	Cycles renders of each module and each finished house type, for your review.
8	Technical report	.md in repo	Measured triangle counts, UV/texel-density checks, texture list, naming, how to use the kit.
9	Source scripts	.py, *tools/*	Everything used to build, bake and export, so it can be re-run.

Not included: collision meshes (to be done at the end, see section 6), LODs (not needed with Nanite), terrain, roads, fields, bridges and other non-house assets in the scene, the village layout solver (*GN_EdoVillage*, not rebuilt unless you ask), and any Unreal project or Blueprint files.

2. Houses covered

The 94 houses are only 17 designs (the generator's parameters match exactly per design; the seed only jitters the footprint). The kit has to cover all 17. The first prototype is **dwellA**, then **kura**, then the thatched types, then the rest.

Design	Count	Bays	Floors	Roof	Extras
kura	14	2x2	2	Tile + pent	-
shopA	13	3x3	1	Tile hip	Shopfront, lanterns
dwellA (prototype 1)	11	4x3	1	Tile hip	Engawa, fence
farmhouse	9	5x3	1	Thatch gable	Engawa, fence
dwellB	9	4x3	2	Tile + pent	Balcony, fence
barn	8	3x4	1	Thatch gable	-
dwellC	7	3x3	1	Tile + pent	Engawa, fence
hutA / hutB / hutC	2 / 3 / 2	3x3 / 4x3 / 3x2	1	Thatch gable	Engawa, fence, lanterns (varies)
restA / restB	3 / 2	5x4 / 6x4	1 / 2	Tile	Shopfront, engawa, balcony, lanterns

Design	Count	Bays	Floors	Roof	Extras
shopB	3	3x4	2	Tile + pent	Shopfront, balcony, lanterns
tea	3	3x3	1	Thatch gable	Engawa, lanterns
row (rowhouse)	2	7x2	1	Tile + pent	-
inn	2	9x6	2	Tile hip	Engawa, balcony, lanterns, gate, stone compound wall

Roof values confirmed from renders in stage 1: 0 = no roof (open frame), 1 = tile hip, 2 = thatch gable, 3 = tile hip with a slightly different ridge cap. Every two-floor house also gets a pent tier between the floors.

3. Module families in the kit

Derived from the architecture of the *GN_EdoHouse* node group: each spawner builds one family. Every family becomes one or more clean, grid-snapped modules.

Family	Pieces (planned)	Replaces generator part
Foundation	Stone pier / footing, stilt post (stilted houses)	EDO_PierSet
Timber frame	Corner post, wall post, tie beam, floor beam, bracing	Timber Frame
Wall bays	Plaster panel, paper/lattice panel, door panel, window variants, corner (one set per bay width, 1 ken)	Wall Modules + Wall Placement (11 box types)
Interior	Partition panels (visible through openings only)	Interior Partitions (6 types)
Tile roof	Hip roof sections (sized by bays), corner/hip ridges, eave edge, ridge cap, end tiles	Main Roof Surface, Smooth Roof
Pent roof	Tile pent tier between floors (2-floor designs)	Pent Roof Surface
Thatch roof	Gable roof sections, ridge log, crossed ridge finials, thick eave edge	Thatched Gable Roof
Entrance	Engawa deck, steps, deck posts and bracing, railings	Entrance (Engawa + Steps)
Balcony	Outer beam, railing sections	Balconies
Shopfront	Noren (cloth) rod, sign arm, sign board, lantern	Shopfront
Plot	Fence segment, fence corner, gate (posts, beams, small roof, lanterns), stone compound wall (inn)	Plot: Fence + Gate
Props	Barrel, firewood stack, jar, rack, rice bale	Props
Weathering	Not geometry: moss, dirt, wear, sag handled in materials and decals	Wear (sag, lean, moss, dirt)

Modules come in sizes matching the existing bay counts (1 to 9 bays) by combining pieces on the grid, not by stretching meshes, so textures keep a constant scale.

4. Technical specification

Item	Specification	Status
Grid	1 ken = 1.818 m per bay (measured from the generator: 1.81 m per bay)	Agreed (from analysis)
Storey height	About 3.14 m per floor (measured)	Agreed (from analysis)
Unreal scale	FBX exported at Unreal scale (1 Blender unit = 1 m = 100 cm), Z up, forward -Y, applied transforms	Proposed
Origins / pivots	Grid-aligned origin at the module's bottom corner or centre-bottom, documented per family	Proposed
Naming	SM_Edo_<family>_<piece>_<variant>, materials M_Edo_<name>, textures T_Edo_<set>_<map>	Proposed

Item	Specification	Status
Meshes	Quads/clean triangles, smooth shading with hard edges only where needed, no n-gons, no hidden interior faces	Proposed
Built for geometry nodes	Every module is authored to be instanced by the new geometry nodes: fixed 1 ken bay width, origin on a grid corner, consistent facing, pieces that tile along X and Y, separate corner and end pieces. The seed only picks between variants; it never changes sizes.	Agreed (new goal)
Supported parameters	Bays X/Y, Floors, Roof style, Shopfront, Engawa, Balcony, Lanterns, Gate, Fence, Props, variant seed. Bay Scale and Raise are fixed (the village only uses 1.0 and 1.12). The 17 existing designs are the guaranteed set; the tested range beyond them will be reported.	Proposed
Placement	Every house keeps its original position: x, y and rotation about Z (0, 90, 180, -90 degrees); z = 0 and scale 1 for all 94 houses, stored in analysis/data/placements.json and used by the manifest.	Agreed
Ground adaptation	The original generator reads the terrain under each house and adapts floor level and stilt length. The kit foundation uses a variable-length post and the new geometry nodes sample the same Ground object, so houses meet the terrain as before.	Proposed (needed to keep houses in place)
LODs	None. Nanite meshes.	Agreed
UVs	UV set 0: tiling/trim layout, consistent texel density across the kit, no overlapping islands except intentional trim reuse. UV set 1 (lightmap) only if you ask for it; not needed with Lumen	Proposed
Vertex colour	Dropped. The current vertex-colour masks are replaced by texture-based materials and decals	Proposed
Materials	Few materials per module (target 2-3) to keep draw/material cost low	Proposed
Seed variation	Fixed modules. Variety comes from module variants and material tint, not geometry jitter	Agreed (my recommendation)

5. Triangle budgets

Part	Budget	Status
One storey (frame, walls, openings, doors)	2,000 - 5,000 triangles	Agreed
Roof (tile or thatch)	up to 10,000 triangles	Agreed
Engawa + steps	up to 3,000	Proposed
Gate	up to 5,000	Proposed
Fence segment	up to 400	Proposed
Prop (barrel, jar, rack, bale, ...)	about 500 each	Proposed
Pent roof tier (2-floor designs)	up to 5,000 each	Proposed (not yet discussed)

A typical one-storey house lands around 12,000 - 22,000 triangles, close to today's median of about 19,000 (current houses: min 5,208, max 81,442, total 1.84 million for 94 houses). The difference is where they go: today the roof tiles are real geometry and the walls are boxes; in the kit, tile and wood detail comes mostly from normal maps and textures on modelled silhouettes. With Nanite these budgets are conservative; hero details (eave edges, bracket ends, ridge ornaments) may use the headroom where it helps the silhouette. I will report the real measured counts per module.

6. Collisions

Agreed: the current scene has no collision meshes and the generator has no collision output, so none can be copied. Collisions are done at the end of the project: simple boxes/convex hulls per module, exported as UCX_ meshes alongside the FBX files. If you want them to match a specific setup from your Unreal project, send a

screenshot or description at that stage.

7. Textures and materials

Photo-based CC0 sets (downloaded by you from Poly Haven / ambientCG into *textures/*; I cannot download them here):

Material	Planned source set	Role
Roof tile	Poly Haven: Roof Tiles (recoloured to dark grey in stage 1); grey_roof_01 kept as colour/weathering reference	Tile roofs
Thatch	Poly Haven: Reed Roof 03 (alternative: Thatch Roof Angled)	Thatched gable roofs
Timber (dark)	Poly Haven: Dark Wooden Planks	Posts, plank walls, fences
Timber (beams)	Poly Haven: Rough Wood	Beams, rails, railings
Plaster	Poly Haven: White Stucco	Wall plaster (warm off-white)
Stone	Poly Haven: Stacked Stone Wall (alternative: Rock Wall 05)	Footings, compound walls
Moss overlay	ambientCG: Moss 001 (licence to be verified)	Moss on roofs, stone, timber

Procedural (made and baked in Blender): paper (shoji), cloth (noren), lanterns, signs, dirt/grime and edge wear, and any gap where the photo sets do not cover a need.

Texture spec	Value
Resolution	4K for the five main materials (tile, thatch, timber, plaster, stone); 2K for minor ones (cloth, paper, lantern, sign, props). Unreal streams lower mips by distance automatically. Status: Agreed (4K); minor-set 2K proposed
Sharing	Shared tileable / trim sets used by every house, not unique 4K textures per house. A full 4K set is roughly 50-65 MB
Maps	BaseColor (sRGB), Normal (DirectX / Unreal convention), packed ORM (Occlusion, Roughness, Metallic), Emissive for lanterns
Format	Sources are the JPG/EXR downloads; outputs are PNG: BaseColor, Normal_DX, packed ORM. Texture preparation script tested in stage 1 on all 8 sets
Licence	CC0 sets are free for commercial use. I will confirm each set's licence on its own page before using it, and list them in the report

Realism: photo-based CC0 sets plus modelled detail give a good semi-realistic to realistic result. It will not match hand-sculpted or fully photoscanned AAA assets up close; final judgement needs your eye in Unreal under real lighting.

8. How the work will be done

Stage	What	Model / effort	You review
0	Analysis of the .blend (done): 17 designs, grid, triangle costs, architecture, contact sheet	Sonnet 5.5	Report in *analysis/*
1	DONE: roof values confirmed, 94 placements recorded, ground behaviour checked, texture preparation tested on all 8 sets	Sonnet 5.5	-
2	Prototype: dwellA (foundation, frame, walls, tile roof, engawa, fence, props), UVs, textures, baked maps, previews, plus a first version of the kit geometry nodes building dwellA and size variants (3x3, 4x3, 5x3) from the modules	Opus 5.5, extra-high	Renders, tri counts, FBX
3	kura: pent roof tier, 2-floor assembly	Opus 5.5, extra-high	Renders
4	Thatched types: farmhouse, barn, huts, tea	Opus 5.5, extra-high	Renders
5	Remaining: shops, restaurants, inn, row, balcony, gate, compound wall	Opus 5.5 / Sonnet 5.5	Renders

Stage	What	Model / effort	You review
6	Bake, pack textures, finish the kit geometry nodes for all 17 designs, export all FBX, build manifest and Unreal import helper	Sonnet 5.5	Files in repo
7	Collisions (UCX), parameter-sweep tests of the geometry nodes, final verification report	Sonnet 5.5	Report

Work is delivered in stages to the repo branch so you can stop or redirect after each one. Stage 2 is the main checkpoint: if the prototype quality is not what you want, we adjust before the other 16 designs.

Budget plan

Using the model prices I have (Opus 5.5: $4 input / $20 output per million tokens; Sonnet 5.5: $2 / $10; cache reads $0.20) and my assumptions about turn size, a full build of roughly 300-500 turns comes to about **$35-60** (this includes roughly 50-100 extra turns, my estimate, for the new geometry nodes) with the split above (Opus at high effort; extra-high, which you chose, costs more and could push toward the upper end or beyond), versus about $18-30 on Sonnet alone and about $50-110 on Opus at max effort. These are estimates, not measurements: the real figure depends on how many render-check-fix loops are needed. I will report spending trends after stage 2 so you can decide whether to continue.

9. What I can and cannot verify

I can verify here	I cannot verify here
- Triangle counts per module and per assembled house - Modules sit on the 1 ken grid, correct dimensions and pivots - UV validity, overlap, texel density consistency - Texture maps generated, sizes, channel packing - FBX exports open and match the .blend (re-import test) - Preview renders (CPU Cycles) - Geometry nodes: parameter sweeps (bays, floors, toggles), no gaps or overlaps, grid alignment, triangle counts vs the original footprints	- How it looks in Unreal under Lumen/Nanite lighting - Nanite behaviour, material compatibility (translucent paper/lanterns), performance - The Unreal import script (no Unreal available) - Collision behaviour in your game - Whether textures downloaded by you match the exact picks (you provide them)

10. Risks and open items

- **No Unreal here.** Everything engine-side is untested until you import it. Paper shoji and lanterns may need opaque/masked + emissive treatment to stay Nanite-friendly; I will flag this per material.
- **Roofs are the hardest part to make modular.** The current roof is a generated surface with curved eaves. A kit roof has to be assembled from pieces that tile for any bay count and still look right at the corners. Stage 2 tests this on dwellA at several sizes; if a fully instanced roof does not hold up, the fallback is roof sections generated by the geometry nodes using the same textures and UVs. I will report which one is used.
- **Thatch texture is not seamless.** reed_roof_03 has a ridge band along its top edge; I will use the straw area for the roof and the band as trim.
- **Realism ceiling.** CC0 photo textures + modelled detail is good but not AAA photoscan quality.
- **Repo size.** Textures and FBX files add up; GitHub rejects any file over 100 MB. Git LFS is recommended for *textures/* and *export/*.
- **Texture downloads depend on you.** This container cannot reach Poly Haven / ambientCG. Licences must be confirmed on each page (Poly Haven states CC0; ambientCG not confirmed by me).
- **Roof style mapping** (Roof 1 / 2 / 3) is inferred from renders and must be confirmed in stage 1.
- **Balcony** showed no change in my test on a single-floor house (probably needs two floors); to be confirmed when building dwellB.
- **Pent roof budget** (5,000) is a new proposal you have not yet seen; tell me if you disagree.
- **Credits.** Costs above are estimates; Opus max effort is not planned because it could use most of the budget.

11. Decisions recorded

Question	Decision
Triangle budget per module	About 2-5K per storey, roof up to 10K. Engawa, gate, fence, props budgeted separately (proposed values above)
Texture resolution	4K, optimized for games; Unreal handles distance mip lowering
Module split	Taken from the node-group architecture (section 3)
LODs / collisions	No LODs (Nanite). Collisions at the end
Seed variation	Fixed modules, variety via variants and material tint
Existing geometry nodes	The kit does not need to work with GN_EdoHouse, which is not modified. It must work with new geometry nodes built from the kit meshes from the start
First prototype	dwellA, then kura
Texture source	Real CC0 photo sets downloaded by the user from Poly Haven / ambientCG, plus procedural materials for the rest
Model plan	Sonnet 5.5 for analysis and mechanical steps; Opus 5.5 at extra-high for the hard modelling stages

Source analysis (measured): see *analysis/KIT_ANALYSIS.md* in the repository.